SAMARTH

CUET PG EXAM





Lecture -1 | Computer Science

Digital Fundamental

Topic : Number System and Conversion Part-1



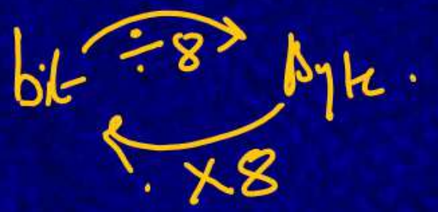
By- Puneet Kaur

Topics *to be covered*

1. Number System and Conversion Part-1



Topic : Number System and Conversion Part-1



Bit	One binary digit
Nibble / Byte	8 bits
Kilobit	1,024 or 2^{10} bits
Kilobyte	1,024 or 2^{10} bytes
Megabit	1,048,576 or 2^{20} bits
Megabyte	1,048,576 or 2^{20} bytes
Gigabit	2^{30} bits
Gigabyte	2^{30} bytes
Terabyte	2^{40} bytes
Petabyte	2^{50} bytes
Exabyte	2^{60} bytes

$1024 = 2^{10}$

1024 ~ 1000

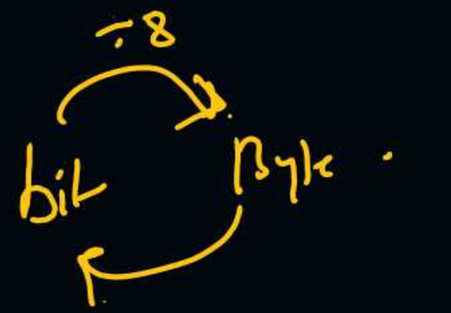
Tb — 1024 x 1024 x 1024 x 1024 bit $= 2^{10} \times 2^{10} \times 2^{10} \times 2^{10} = 2^{40} \sim 1000000000000$

Gb — 1024 x 1024 x 1024 bit $= 2^{10} \times 2^{10} \times 2^{10} = 2^{30} \sim 1000000000$

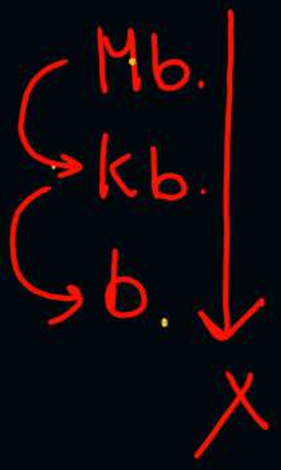
Mb — 1024 x 1024 bit $= 2^{10} \times 2^{10} = 2^{20} \sim 1000000$

kb — 1024 bit $= 2^{10} \sim 1000$

bit — 0, 1



Q: 10 Mb = ? KB



10×1000000

$= 10000000$ bit

$= \frac{10000000}{8}$ Byte

$= 1250000$ Byte.

$= \frac{1250\cancel{000}}{1\cancel{000}} =$ 1250 KB



Q: 100 kB = ? Gb

Gb	↑
Mb	
kb	
bit	

kB	
Byte	↓

① 100×1000 Byte.

$= 100000$ Byte.

② Byte $\xrightarrow{\times 8}$ bit

$= 100000 \times 8$

$= 800000$ bit

$= \frac{8\cancel{00000}}{1000\cancel{00000}} = \frac{8}{10000} = \frac{8}{10^4} = 8 \times 10^{-4}$

$= 0.0008$ Gb

Topic : Number System and Conversion Part-1

Binary, octal, decimal, and hexadecimal are all number systems used in digital electronics and modern computing:

Decimal $( )_{10}$ 0 to 9

- Uses base 10 and the place values are based on the powers of 10. Any number can be represented using the digits 0–9.

Binary $( )_2$

- Uses base 2 and each digit can only be a 0 or 1.

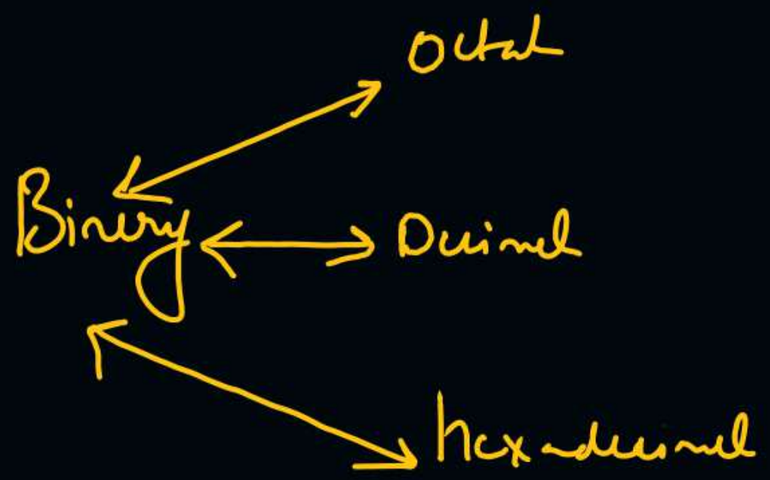
Octal $( )_8$

- Uses base 8 and the value of each place is based on the powers of 8. Any number can be represented using the digits 0–7.

Topic : Number System and Conversion Part-1

Hexadecimal

- Uses base 16 and the value of each place is based on the powers of 16. Any number can be represented using the digits 0–9 and the characters A, B, C, D, E, and F.
- To convert between bases other than decimal, it's usually necessary to first convert the number to base 10, and then convert that number to the second base. However, it's easy to convert directly between binary and octal, and between binary and hexadecimal.



(1) Binary to Decimal

$(\quad)_2$ to $(\quad)_{10}$

$(\text{Any no})^0 = 1$

$2^{-1} = \frac{1}{2} = 0.5$

$2^{-2} = \frac{1}{2^2} = \frac{1}{4} = 0.25$

$2^{-3} = \frac{1}{2^3} = \frac{1}{8} = 0.125$

Ex 10110

$1 \times 2^0 = 0$

$1 \times 2^1 = 2$

$1 \times 2^2 = 4$

$0 \times 2^3 = 0$

$1 \times 2^4 = 16$ +

22

$10110 \cdot 101$

22 . 625

0.625

$.101$

$1 \times 2^{-1} = 0.5$

$0 \times 2^{-2} = 0$

$1 \times 2^{-3} = 0.125$ +

② Decimal to Binary.

Ex $(56)_{10} = (?)_2$

2	56	
2	28	0
2	14	0
2	7	0
2	3	1
	1	1

$(111000)_2$

56 . 70
Lhs Rhs

.10110

$.70 \times 2 = 1.40$

$.40 \times 2 = 0.80$

$.80 \times 2 = 1.60$

$.60 \times 2 = 1.20$

$.20 \times 2 = 0.40$

⋮

③ Binary to Octal.

$()_2$ to $()_8$

Ex: 001011010

001 011 010

1 3 2

Octal (0-7)

000 — 0
001 — 1
010 — 2
011 — 3
100 — 4
101 — 5
110 — 6
111 — 7

Ex: 001011010.101000

1 3 2 . 5 0

132.50

④ Octal to Binary

$()_8$ to $()_2$.

Ex: 7 4 6

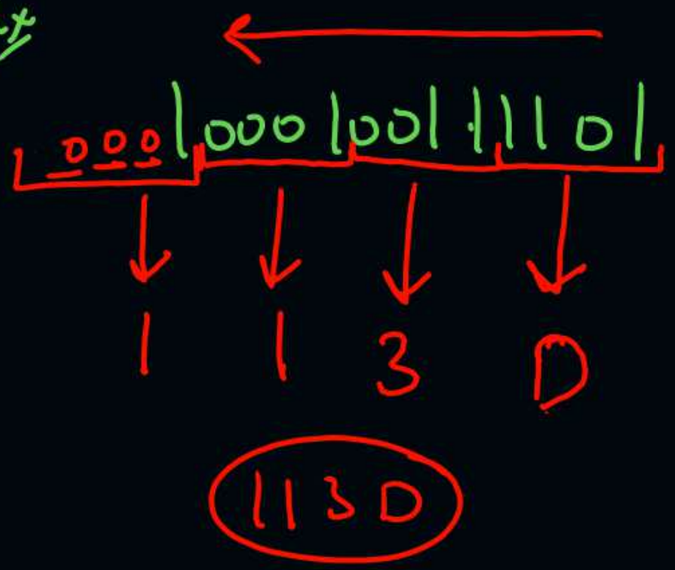
111 100 110

Ex: 746.23

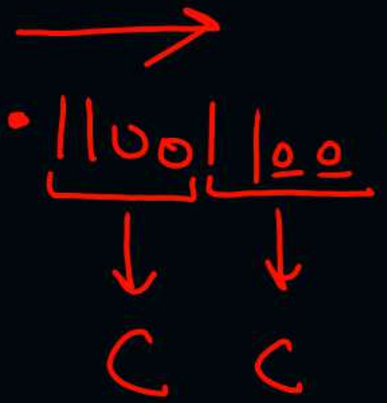
111 100 110 . 010 011

Binary to Hexadecimal

Ex



Ex



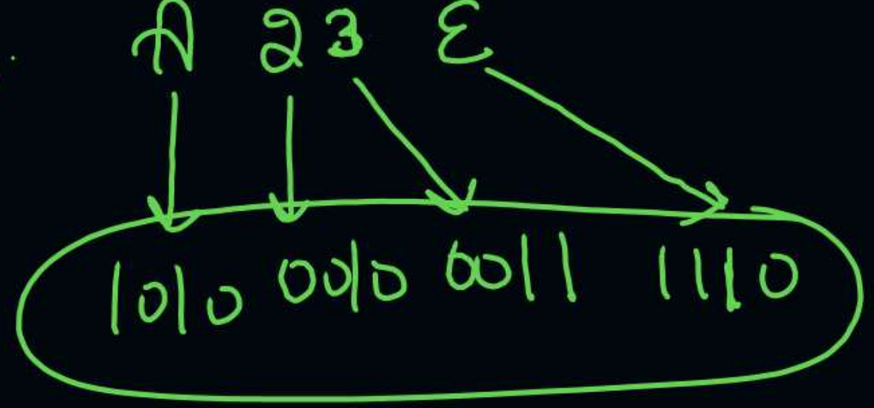
Hexadecimal

0-9, A-F.

0	0	0	0	— 0	
0	0	0	1	— 1	
0	0	1	0	— 2	
0	0	1	1	— 3	
0	1	0	0	— 4	
0	1	0	1	— 5	
0	1	1	0	— 6	
0	1	1	1	— 7	
1	0	0	0	— 8	
1	0	0	1	— 9	
1	0	1	0	— 10	A
1	0	1	1	— 11	B
1	1	0	0	— 12	C
1	1	0	1	— 13	D
1	1	1	0	— 14	E
1	1	1	1	— 15	F

Hexadecimal to Binary

Ex



Ex A 2 3 E . 8 B

1010 0010 0011 1110 . 1000 1011

Topic : Number System and Conversion Part-1

Here are some ways to convert between number systems:

- Decimal to binary: Divide the number by 2.
- Decimal to hexadecimal: Divide the number by 16.
- Decimal to octal: Divide the number by 8 and write the remainders in reverse order.

Topic : Number System and Conversion Part-1

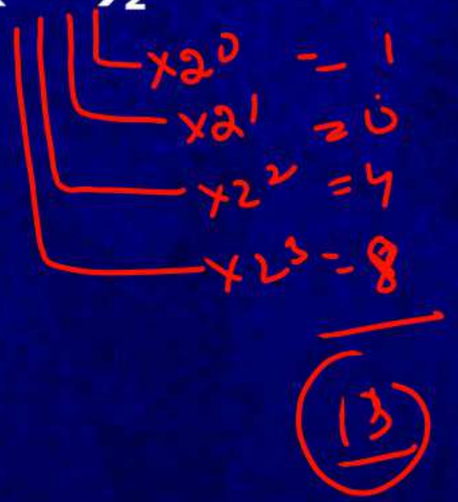
#Q. Convert the binary number $(1101)_2$ into a decimal number.

A. 12

B. 13

C. 14

D. 15



Topic : Number System and Conversion Part-1

Ans. B

Topic : Number System and Conversion Part-1

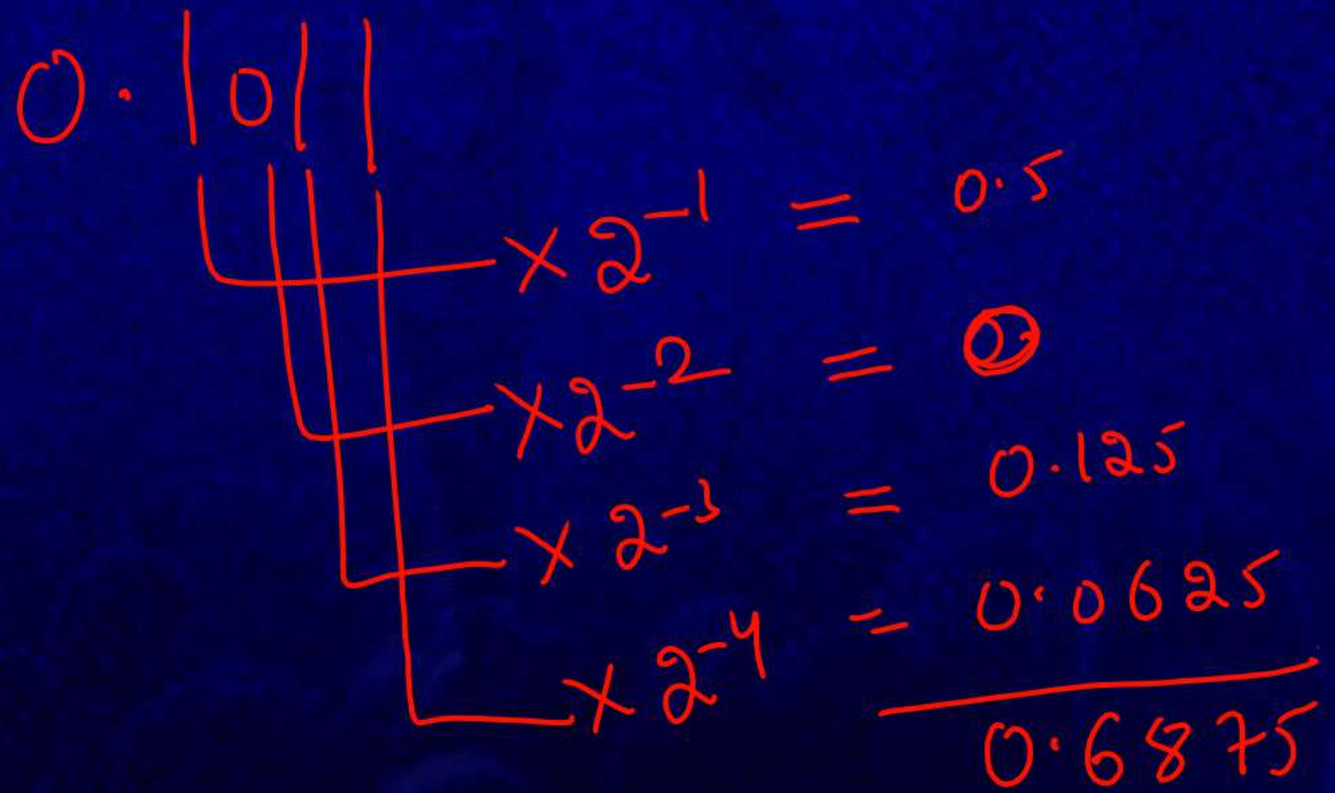
#Q. Convert 0.1011 to decimal.

A. 0.6875

B. 0.6857

C. 0.5867

D. 0.7865



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Ans. A

Topic : Number System and Conversion Part-1

11011.1001

#Q. Convert decimal number 27 to binary.

A. 110011

B. 11001

C. 11011

D. 11101

Ex 27.62
1001

2	27
2	13 1
2	6 1
2	3 0
	1 1

.62 x 2 = 1.24

.24 x 2 = 0.48

.48 x 2 = 0.96

.96 x 2 = 1.92

.92 x 2

Topic : Number System and Conversion Part-1

Ans. C

Topic : Number System and Conversion Part-1

1010.01

#Q. Convert decimal number 10.25 to binary.

A. 1010.10

B. 1010.01

C. 1010.00

D. 1010.11

.25 X 2 = 0.50

.50 X 2 = 1.00

.00 X 2 = 0

Topic : Number System and Conversion Part-1

Ans. B

Solution:

$(10.25)_{10} = (?)_2$

For 10:

2	10
2	5 --- 0
2	2 ---1
2	1 --- 0
2	0 --- 1

Radix Point

= 1010

For .25:

2 * 0.25 = 0.50

2 * 0.50 = 1.00

= 01

Result is: $(1010.01)_2$

Topic : Number System and Conversion Part-1

#Q. Convert Binary number 11101011 to octal.



A. 352

B. 253

C. 353

D. 453

Topic : Number System and Conversion Part-1

Ans. C

Topic : Number System and Conversion Part-1

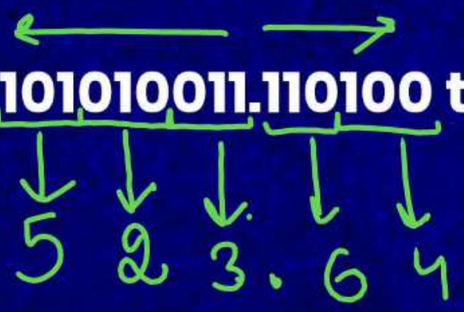
#Q. Convert binary number 101010011.110100 to octal.

A. 523.64 ✓

B. 533.64

C. 543.65

D. 533.56

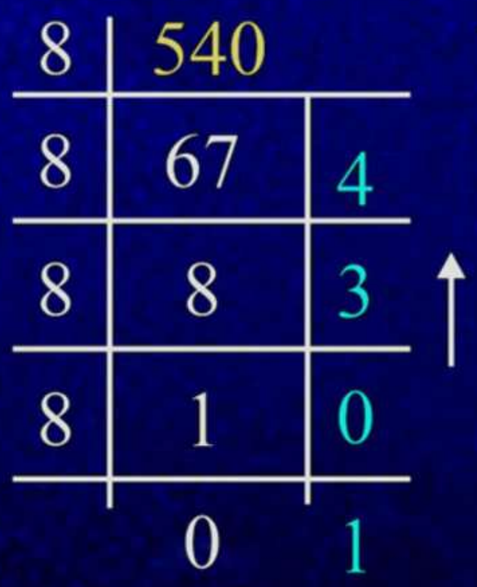


Topic : Number System and Conversion Part-1

Ans. A

Topic : Number System and Conversion Part-1

Ans. C



Decimal to Octal

$(540)_{10} = (1034)_8$

Topic : Number System and Conversion Part-1

Ans. B

29.30

8 |	29	
8	3	5 ↑
	0	3

$(29.30)_{10} = (35.23146)_8$

$0.30 \times 8 = 2.4$
$0.40 \times 8 = 3.2$ ↓
$0.20 \times 8 = 1.6$
$0.60 \times 8 = 4.8$
$0.80 \times 8 = 6.4$
$0.40 \times 8 = 3.2$] Stop
$0.20 \times 8 = 1.6$] Repeated
$0.60 \times 8 = 4.8$]
$0.80 \times 8 = 6.4$]

2 Mins Summary

Number system.

↳ bit | byte

↳ Binary ↔ octal
Binary ↔ Decimal
Binary ↔ hex.

H.W.

YT

By Puneet Kaur

Story of bit & Byte

THANK YOU